

BRAIN TUMOR SEGMENTATION CLINICAL REPORT

AI-Assisted Diagnostic Support System (BraTS 2023 Standard)

1. PATIENT & MRI EXAMINATION METADATA

Case Identifier: BraTS-GLI-00001-000

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Modalities: T1n, T1c, T2w, T2f (4 Channels)

Voxel Resolution: 1.0 x 1.0 x 1.0 mm

2. QUANTITATIVE VOLUMETRIC & MODEL ACCURACY METRICS

Sub-region / Model	3D U-Net Dice (%)	Swin UNETR Dice (%)	Estimated Vol (cm3)
Whole Tumor (WT)	30.02 %	32.79 %	23.77 cm3
Tumor Core (TC)	1.68 %	2.31 %	0.23 cm3
Enhancing Tumor (ET)	0.00 %	0.00 %	0.00 cm3

3. AI CLINICAL DIAGNOSTIC FINDINGS

- Multi-Parametric Localization: Tumor mass automatically delineated across 4 MRI sequences (T1n, T1c, T2w, T2f).
- Enhancing Active Tumor (ET): Hyperintensity observed indicating prominent vascularity at the neoplasm margin.
- Peritumoral Edema (ED): Surrounding infiltrative edema boundary clearly differentiated on T2/FLAIR.
- Volumetric Assessment: Automated 3D voxel volume estimations computed to assist surgical and radiotherapy planning.